Motion with constant acceleration along a straight line

1. If a particle has initial velocity 2.4 ms^{-1} and accelerates at a constant rate of 1.3 ms^{-2}, what is its velocity 4 seconds later?

2. If the particle has initial velocity u ms^{-1} and accelerates at a constant rate of a ms^{-2}, what is its velocity v ms^{-1}, t seconds later?

3. A particle with initial velocity 5 ms^{-1} accelerates uniformly over 8 seconds until it has velocity 23 ms^{-1}.
   (a) Draw a velocity-time graph. (b) Hence calculate the displacement from the starting position.

4. If in general the particle with initial velocity u ms^{-1} accelerates uniformly over t seconds to reach a velocity of v ms^{-1}, what is its displacement, s m, from its starting position?

5. If the particle has initial velocity 2.4 ms^{-1} and accelerates at a constant rate of 1.3 ms^{-2}, what is its displacement from its starting position 4 seconds later?

6. If the particle has initial velocity u ms^{-1} and accelerates at a constant rate of a ms^{-2}, what is its displacement s m from its starting position t seconds later?

7. Use your results from questions 2 and 4, eliminating t, to obtain a formula involving v, u, a and s.

8. Use your results from questions 2 and 4, eliminating u instead, to obtain a formula involving v, t, a and s.